

LAYAN KAUSHIK

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EDUCATION

Duke University, Durham, NC, USA

Aug 2022 – May 2024

M.S. Economics and Computation

GPA: 3.91/4.0

Courses: Theory and Algorithm of Machine Learning, Probabilistic Machine Learning, Elements of Deep Learning, Design and Analysis of Algorithms, Advanced Econometrics, Numerical Analysis, Graph Network Analysis, Algorithmic Game Theory.

Indian Institute of Technology, Roorkee, India

Jul 2016 – May 2020

B.Tech. Civil Engineering, First Division with Distinction

GPA: 8.726/10

Courses: Computer Programming and Numerical Analysis, Engineering Mathematics, Probability and Statistics, Data Mining, Linear Optimization, Graduate Econometrics.

PUBLICATIONS

“Refining FEMA’s NFIP V2 Open Flood Insurance Claims Dataset: Comprehensive Enhancements through Data Augmentation, Feature Engineering, Anomaly Detection and Multiple Imputation”

American Geophysical Union (AGU) Conference 2023, ([Publication Link](#))

Flood damage modeling faces critical data limitations. We enhanced FEMA’s NFIP V2 dataset (2.5M+ claims, 1978-2023) by integrating Census and meteorological data, developing novel geographic units for precise geo-location, and applying machine learning methods to detect and correct systematic errors. Multiple imputation addressed missing data. Both imputed and non-imputed datasets are released to advance flood risk analysis and damage prediction research.

WORKING PAPERS

“SurveyBalancer: Statistical Survey Reweighting Package”

Amazon, January 2025 - Present

Survey non-response bias threatens data reliability at scale, particularly for product strategy decisions. SurveyBalancer ensures robust bias correction through machine learning-based propensity score estimation and statistical reweighting methods. It implements leave-features-out cross-validation and distribution matching techniques. Deployed across Amazon’s survey research teams, it enhances survey validity for product decisions affecting millions of customers.

“A Large-Scale Test of Difference-in-Differences and Common Practices with Experimental Data”

Amazon, June 2025 - Present

Systematic validation against experimental ground truth remains limited for Difference-in-Differences (DiD) methods. We conduct large-scale validation using approximately 500 Alexa experiments, comparing DiD estimates against experimental benchmarks and assessing estimator performance under realistic parallel trends violations, varying pre-treatment periods, treatment timing patterns, and covariate adjustment strategies, producing empirical guidance for practitioners.

“Strategy-Proof Mechanism Design for Multi-stage Tournaments”

Duke University, June 2024 - Present

Multi-stage tournaments face strategic manipulation when players underperform to secure favorable future matchups. We propose a strategy-proof reward mechanism using scaled bracket-based incentives and dynamic Elo-style rating updates. Mathematical proofs establish incentive compatibility, validated through Monte Carlo simulations. Applied to real-world formats, our framework identifies critical design flaws and provides incentive-aligned alternatives for tournament organizers.

WORK EXPERIENCE

Economist Apprentice, Amazon, Seattle, US

Jun 2024 - Present

- Working alongside **Professor Alexandre Mas** and **Principal Economist Dr. Mingyu Chen** at the intersection of economics and statistics—specializing in survey methodology, measurement science, and causal inference frameworks.

Product Data Scientist, Mastercard, India

Aug 2020 - Jun 2022

- Designed **geospatial allocation algorithm** to transform Mastercard’s 150m grid-level transaction data into census geographies (zip codes, tracts), using hybrid weighting of area overlap and transaction density to handle boundary conflicts.

RESEARCH ASSISTANT EXPERIENCE

Applying NLP to Uncover Insights in Influencer Marketing, Guided by Prof. Tong Guo, The Fuqua School of Business, Duke University *Jan 2024 - May 2024*

- Analyzed 1.29 million ADHD-related TikTok videos using NLP and unsupervised ML, classifying each as Ad or non-Ad.
- Applied topic modeling to identify behavioral patterns in user engagement, focusing on uncovering prevalent themes and trends within ADHD-related discussions on TikTok.

Treatment Effect Estimation With Measurement Error in Covariates and Repeat Measurements, Guided by Prof. Michael Pollmann, Duke University *May 2023 - Jan 2024*

- Derived and implemented unbiased estimation methods in R, including polynomial errors-in-variables models (Hausman et al.) and treatment effect estimation with noisy conditioning variables (Nagasawa).
- Collaboratively developed an optimized R package with extensive simulations on Duke's Computer Cluster and implemented cross-validation code for CoCoLasso regularization, addressing estimation with measurement errors.

Addressing User Activity Bias in Bipartite Graph Ranking, Guided by Prof. Xiaobai Sun, Duke University *Sep 2023 - Dec 2023*

- Enhanced the BiRank algorithm for bipartite graph rankings in recommendation systems, incorporating user activity bias mitigation through a novel regularization technique and developing a unique fairness metric combining activity rank balance and Gini coefficient.
- Executed extensive experimental analysis on synthetic datasets, demonstrating significant fairness improvements in rankings, particularly for power-law distributed graphs.

Ride Sharing and Tipping Behavior in Chicago from 2018 to 2023, Guided by Prof. Jian Pei, Duke University *Jan 2023 - May 2023*

- Analyzed 300M+ Chicago ride-sharing records (2018-2023) using regression, time-series analysis, and hypothesis tests in python language; identified statistically significant patterns, including a 33% ride drop post-pandemic and 1.5% tipping increment, reflecting socio-economic trends.

Spatial Distribution of Moisture Content in Cement Mortar Structures using Radial Basis Functions, Guided by Prof. Bishwajit Bhattacharjee, IIT Delhi *May 2018 - Aug 2018*

- Research to measure moisture content in cement mortar required to obtain durability and service life of the structure.
- Interpolated using Gaussian and inverse multiquadric radial basis method to determine moisture content for missing data.

TEACHING EXPERIENCE

Graduate TA, CS230: Discrete Math for CS, Duke University *Jun 2023 - Aug 2023*

Graduate TA, CS216: Everything Data, Duke University *Jan 2023 - May 2023*

Undergraduate TA, Computer Programming in C++, IIT Roorkee *Jan 2018- May 2018*

HONORS AND AWARDS

Dean's Research Award for Master's Students, Duke University *2023*

U.S. Army Corps of Engineers Institute for Water Resources Research Grant, Washington DC *2023*

Duke Graduate Working Group on Global Issues Award, Duke University *2023*

Duke Economics Master's Scholar Award, Duke University *2022-24*

K. C. Mahindra Scholarship for Post Graduate Studies Abroad, K. C. Mahindra *2022*

IASc-INSa-NASI Summer Research Fellowship, Indian Academy of Sciences *2018*

Top 3 in State Mathematics Olympiad, Homi Bhabha Centre for Science Education, India *2014*

SKILLS

Programming Languages: Python (scikit-learn, PyTorch, pandas, statsmodels, econml, geopandas), SQL, R, C++

ML Frameworks: PyTorch, JAX

Software Packages: Stata, Tableau, MS Excel, LaTeX

Spoken Languages: English (fluent), Assamese (native speaker), Hindi (fluent)

VOLUNTEER WORK AND OTHER ACTIVITIES

Assistant Badminton Coach, Duke University *Aug 2022 - May 2023*

Student Mentor, Student Mentorship Program, IIT Roorkee *Jul 2018 - May 2019*